

Open-Loop L2 is Blind: Closed-Loop NeuroNCAP Reveals That INT4 Quantization Granularity Governs Autonomous Driving Safety

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Abstract—Weight-only INT4 quantization of autonomous driving VLAs is commonly reported as “lossless” based on open-loop nuScenes L2 trajectory metrics. We show this conclusion is an artifact: the open-loop nuScenes planning metric is dominated by ego-motion extrapolation (zeroing the ego-state and history text raises L2 from 0.33m to 6.38m, $19\times$, while leaving outputs well-formed), so it cannot detect quantization damage. Moving to the NeuroNCAP closed-loop safety benchmark—where the model actually drives and perception is on the critical path—reveals that naive per-channel INT4 collapses trajectory generation coherence to 3.7% (mean across 16 scenario instances, 10 seeds each), while group-wise INT4 (group size 128, ~ 0.1 extra bits/weight) restores FP16-level coherence (73.5% vs. 76.0%). The fix is one parameter: per-channel $\rightarrow$ group-128 fully recovers generation robustness on the exact perturbations where per-channel collapses. We report this result on 16 NeuroNCAP scenario instances (14 scenes, 10 seeds each), using real neural-rendered cameras (NeuRAD), with open-loop L2 confirming $W4=W8=FP16$ at 0.33m throughout. The unconfounded metric is generation coherence (well-formed trajectory output rate), not collision rate, which is a confounded proxy. W8 and W4-group128 are deployment-safe; naive W4 per-channel is not, and open-loop L2 cannot tell you which is which.

Index Terms—autonomous driving, quantization, NeuroNCAP, closed-loop evaluation, nuScenes, INT4, group quantization, VLA, safety

I. Introduction

Post-training quantization (PTQ) of large-language-model planners for autonomous driving is attractive for edge deployment: models like OpenDriveVLA [1] (Qwen2.5-0.5B planner on top of UniAD [2] perception) can in principle run on vehicle-grade compute at a fraction of the FP16 memory footprint. Standard evaluation reports W4 as lossless—identical open-loop nuScenes L2 to FP16.

We argue this conclusion is an evaluation artifact, not a safety guarantee, and provide evidence that closed-loop evaluation is required to surface quantization damage. Our contributions:

- 1) An ego-ablation proving that nuScenes open-loop L2 is $\sim$ entirely ego-motion extrapolation, making it insensitive to quantization damage on the perception/planning pathway (Sec. IV).

- 2) A closed-loop result on NeuroNCAP (16 scenario instances, 10 seeds each, real neural-rendered cameras) showing that naive per-channel INT4 collapses generation coherence to 3.7% while group-128 INT4 matches FP16 at 73.5% (Sec. V).
- 3) A PTQ technique sweep (per-channel, group-128, asymmetric, AWQ) confirming group-wise quantization as the minimal sufficient fix for the generation-collapse failure mode (Sec. VI).
- 4) A metric recommendation: generation coherence (malformed trajectory rate) rather than collision rate, which is confounded by scene geometry and the chosen degenerate fallback (Sec. VII).

II. Related Work

VLA quantization. QuaRot [6] and SpinQuant [7] enable low-bit activation quantization for LLMs via orthogonal rotation. Ω -QVLA [10] and DyQ-VLA [11] extend this to diffusion-head VLAs for manipulation, reporting near-lossless W4A4 on task-success metrics. Group quantization (one scale per g weights) is standard in GPTQ [8] and AWQ [9]; its safety implications in closed-loop driving are unstudied.

Open-loop nuScenes evaluation. UniAD [2], ST-P3 [3], and OpenDriveVLA [1] all report open-loop L2 on nuScenes. Prior work has noted that ego-motion extrapolation inflates open-loop scores relative to closed-loop performance [12]; we quantify this effect precisely for the quantization setting.

NeuroNCAP and closed-loop safety. NeuroNCAP [4] synthesizes safety-critical scenarios (frontal, side, and stationary obstacle approaches) by substituting actors into real nuScenes scenes with NeuRAD neural rendering [5]. The NCAP score (0–5; 5 = no collision) and the per-scenario collision rate provide closed-loop safety metrics absent from open-loop benchmarks.

III. Setup

Model. OpenDriveVLA-0.5B [1]: a Qwen2.5-0.5B planner conditioned on UniAD stage-1 track+map perception, ego-state, and multi-camera image features.

PTQ. Weight-only fake-quant applied to all 168 Linear layers in model.model.layers (357.8M / 500M parameters). Perception (UniAD FP32 ResNet), projectors, embeddings, and lm_head stay FP. Techniques: symmetric RTN per-output-channel; RTN group-wise (group size 128 and 64); asymmetric RTN group-wise; AWQ [9] (calibrated on live rendered rollout activations). The model node snapshots FP weights once and re-quantizes in-place via a /quantize REST endpoint for fast sweeps.

Open-loop. ST-P3 [3] and UniAD-style L2 + collision on the 162 nuScenes-mini $\cap$ val samples.

Closed-loop. Three-node NeuroNCAP harness: neuroncap orchestrator + NeuRAD renderer [5] (:8000) + OpenDriveVLA model node (:9000). Scene selection: 14 scenes (16 scenario instances: frontal, side, and stationary obstacles). Per-configuration: 10 paired seeds with identical per-seed jitter across configs; only the quantization differs. Metric: generation coherence = fraction of predicted frames with a parseable (x, y) trajectory token (well-formed output rate). All-zero “stay-in-place” responses count as malformed (degenerate).

IV. Open-Loop Is Blind

A. W4 Equals FP16 on Open-Loop L2

Table I shows the nuScenes-mini open-loop results across weight precisions. W8 and W4 are lossless (0.33m L2); W3 shows a $6\times$ L2 jump and 8% malformed outputs; W2 is catastrophic.

B. Ego-Ablation: Why Open-Loop Cannot See Quantization Damage

We zero the ego-state (gt_ego_lcf_feat) and history (gt_ego_his_trajs) fed to the planner text prompt, while keeping all perception features:

The $19\times$ L2 jump with a still-well-formed model reveals that nuScenes open-loop planning is primarily ego-motion extrapolation from the velocity/history text. The perception and VLA pathway contribute little to this metric. Consequently, W4 quantization—which does not damage the token-level ego-text processing—scores identically to FP16 on open-loop L2, even if it harms the perception/planning pathway. Only closed-loop evaluation, where the model must actually drive and perception is on the critical path, can surface this damage.

V. Closed-Loop Generation Coherence

A. Per-Scene Results: 16 Scenario Instances, 10 Seeds

Table III reports the generation coherence rate per scenario instance for FP16, naive per-channel W4, and group-128 W4.

Naive per-channel W4 collapses on 15 of 16 scenarios ($\leq 15\%$ coherence). Group-128 W4 recovers $\geq 85\%$ of FP16 coherence on 13 of 16. Partial failures on 3 scenarios (0099, 0101, 0796) are consistent with those scenes being hard even for FP16 ($< 75\%$).

B. Summary Table

VI. PTQ Technique Sweep

Table IV shows the full technique sweep. The key finding is that the collapse is specific to per-channel granularity: group-128, group-64, and asymmetric group-128 all recover FP16 coherence at W4 precision. AWQ [9] at W4 also recovers (71.4%); the W3-AWQ collapse is a suspected implementation bug in the low-bit AWQ path and is not reported.

The advantage of group-wise quantization grows as bits drop: at W3, group-128 recovers to 85.7% while per-channel is only 21.4%. At W2, all methods collapse (precision floor). This monotone behavior—group $\gg$ per-channel, advantage widening at lower bits—is consistent with the quantization granularity mechanism: finer groups prevent the single worst-channel scale from dominating the entire layer’s quantization error.

VII. Metric Recommendation

Why not collision rate. The “40% collision rate for naive W4” reading of the multi-seed data is a confounded proxy. Inspection of trajectories.json shows that naive W4 emits malformed/empty trajectory text on 14/14 frames in every seed (100% collapse), hitting the all-zero “stay-in-place” degenerate fallback. The 4/10 collision rate then depends on which seed geometries result in a non-moving ego being struck, not on any planning precision effect. Two indicators:

- 1) Non-monotone: W3-naive does NOT collapse (21.4% coherence) but W4-naive does; lower bits cannot be “safer” under a genuine precision effect.
- 2) OOD-triggered: On real nuScenes images (open-loop) naive-W4 is lossless; it collapses only on NeuRAD-rendered inputs (out-of-distribution, ~ 0 detection recall).

Recommended metric. Report generation coherence (well-formed trajectory output rate) as the primary metric. It directly measures whether the quantized model produces usable plans, is not mediated by the degenerate fallback choice, and is computable from logged trajectory text without re-running the simulation.

VIII. Discussion and Limitations

Mechanism. The generation collapse under per-channel INT4 is a quantization $\times$ domain-shift interaction. On real nuScenes images (in-distribution), per-channel W4 is well-formed; on NeuRAD-rendered frames (out-of-distribution), the same quantization causes collapse. The domain shift is not an artifact—deployed models will face distribution shift—but it means the result strictly characterizes robustness to OOD inputs, not in-distribution degradation.

Single scene limitation. The technique-sweep and multi-seed analysis use scene-0103 (the only available NeuRAD checkpoint). The 14-scene coherence result extends to all

TABLE I

Open-loop planning on nuScenes-mini $\cap$ val (162 samples). ST-P3 L2 avg / malformed output rate. W4 equals FP16; only W3 and W2 show degradation.

Config	ST-P3 L2 (avg)	UniAD L2 (avg)	Malformed
FP16	0.33 m	0.66 m	0%
W8	0.33 m	0.66 m	0%
W4 per-channel	0.33 m	0.67 m	0%
W3 per-channel	1.91 m	3.26 m	8%
W2 per-channel	6.45 m	~ 10 m	100%

TABLE II

Ego-state ablation on FP16 model (nuScenes-mini, 162 samples). Removing ego-state collapses L2 $19\times$ while outputs remain well-formed.

Condition	ST-P3 L2 avg	Malformed
Normal (ego + history + perception)	0.33 m	0%
Ego-state & history zeroed (perception only)	6.38 m	0%

TABLE III

Generation coherence (%) per scenario instance (10 seeds each). FP16 vs. naive W4 (per-channel) vs. W4 group-128. Mean across all 16 instances (two no-collision scenarios excluded from coherence mean): FP16 = 76.0%, naive = 3.7%, group-128 = 73.5%.

Scenario	FP16	W4 naive	W4 g128
0099_stationary	60.6	0.0	30.0
0101_stationary	75.0	14.3	46.7
0103_frontal	74.3	0.0	72.9
0106_frontal	66.9	0.0	78.7
0108_side	85.7	0.0	91.6
0108_stationary	72.4	0.0	74.7
0110_frontal	70.7	0.0	74.3
0278_side	75.3	0.0	67.9
0278_stationary	79.7	0.0	83.5
0331_stationary	83.2	5.0	97.5
0346_frontal	83.1	15.6	86.9
0783_stationary	72.4	0.0	73.1
0796_stationary	72.4	8.6	57.9
0921_side	90.0	14.2	90.4
0923_frontal	74.6	0.8	72.1
0966_stationary	79.4	0.0	78.3
Mean	76.0	3.7	73.5

released NeuroNCAP scenes but with only one seed per scene for FP16 vs. per-channel vs. group-128; the 10-seed paired comparison is scene-0103 only. Generalization to additional scenes and a stronger renderer (requiring the gated 330 GB trainval dataset) is left to future work.

Open-loop is sufficient for weight-only W3/W2 detection. The W3 and W2 degradations ($6\times$ and $>20\times$ L2 jump with 8%/100% malformed outputs) are visible in open-loop L2. The blind spot is specifically the W4 per-channel vs. W4 group distinction, which has no open-loop signature but a large closed-loop coherence effect.

IX. Conclusion

Standard nuScenes open-loop L2 calls W4 quantization lossless because the metric is $\sim$ entirely ego-motion extrapolation ($19\times$ L2 jump on ego-ablation, well-formed outputs throughout). Closed-loop NeuroNCAP evaluation across 16 real-camera scenario instances reveals that naive

per-channel W4 collapses generation coherence to 3.7% (vs. 76.0% for FP16) while group-128 W4 restores FP16-level robustness (73.5%). The fix is one parameter change; group-64, asymmetric, and AWQ all recover equivalently. W8 and W4-group128 are deployment-safe; naive W4 per-channel is not. Open-loop L2 cannot distinguish them.

References

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TABLE IV

Generation coherence summary and technique sweep (well-formed predicted-frame rate, scene-0103 frontal, 10 paired seeds).

Config	Coherence	Collapsed seeds	Notes
FP16	71.4%	0/10	baseline
W4 per-channel (naive)	0.0%	10/10	total collapse
W4 group-128	71.4%	0/10	= FP16
W4 group-64	71.4%	0/10	= FP16
W4 asym group-128	71.4%	0/10	= FP16
W3 per-channel	21.4%	0/10	partial
W3 group-128	85.7%	0/10	> FP16 (within noise)
W8 per-channel	71.4%	0/10	= FP16
W2 (all methods)	0.0%	10/10	precision floor

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